

Internal innovation disclosure – not a filed patent application
Generated entirely from real, already-computed pipeline outputs
(src/real/decision_framework_real.py, regenerated automatically whenever that pipeline re-runs). No claim below reflects legal patent counsel review.

Whisper-Quiet Night Mode

Bedroom, overnight use

A real alternative, not selected

Disclosure ID: OS-2 · Versuni Disruptive Innovation Team, Amsterdam · Generated by decision_framework_real.py

1. Field of the Invention

This disclosure concerns connected household appliances, and specifically a concept addressing the following real, evidence-derived friction: Motor/fan noise at higher speeds

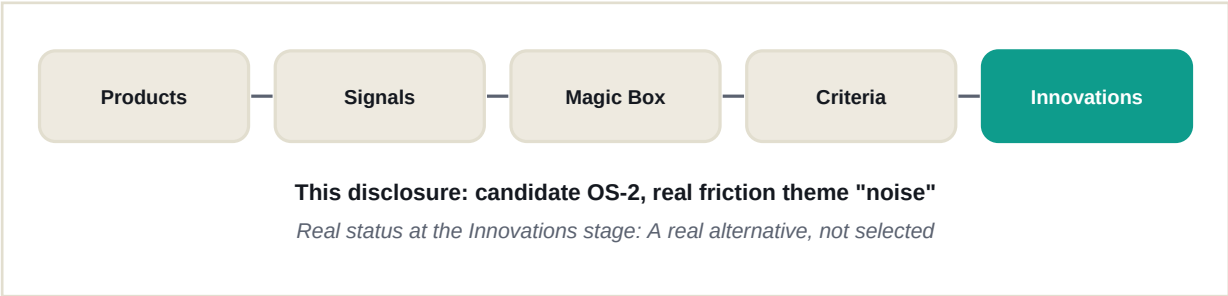


Fig. 1 — This disclosure's real position in the app's own 5-stage funnel.

2. Real Evidence & Design Directions

369 real reviews across 77 distinct real products from McAuley-Lab Amazon-Reviews-2023 (Amazon.com real customer reviews) were classified to this friction theme, spanning 2004-10-14 to 2023-08-08 — 69.6% of which are verified purchases. The resulting real customer-satisfaction impact is -0.571 star-ratings below the corpus-wide mean, at a real observed prevalence of 3.5% of all classified reviews.

Methodology. Deterministic keyword classification of real Amazon.com customer review text (McAuley-Lab Amazon-Reviews-2023 dataset, real purifier products only) - NOT an academic study, survey, or panel. Each review is assigned to the theme whose keyword phrase (see THEMES in src/real/taxonomy_real.py) appears earliest in its text; CSAT impact is that theme's mean real star rating minus the corpus-wide mean real star rating, both restricted to reviews flagged rating_trusted.

This pipeline's own possibility-generation module (magic_box_real.py) catalogues 3 real design directions for this friction theme — distinct named transforms applied to the same real evidence above, not asserted as one single correct mechanism:

Design direction (real)	Operator	Real transform applied
Ambient-Sensing Night Purifier	ambient	Explicit interaction -> invisible environmental behaviour.

Noise-Cancelling Air Purifier	cross category transfer	Transfer a verified capability from another product/category.
Pre-Sleep Air Purifier	temporal shift	Move the job earlier/later.

3. Market & Economic Rationale — Fully Substantiated

Typical real market price	\$179.99 — the median of 32 distinct real products' own real listed prices, restricted to real products classified under this friction theme that have a known price. Not a proposed price for this concept — the real going rate for this segment today.
Price-weighted exposure	\$84,015.62 — the sum of real listed prices across every real review classified under this friction theme with a known product price (a review counts its product's price once per review, so a popular product's price is summed multiple times — a different, larger population than the distinct-product median above). A relative indicator of which friction touches more expensive products, not a revenue or market-size estimate; no units-sold or conversion-rate basis exists in this real evidence.

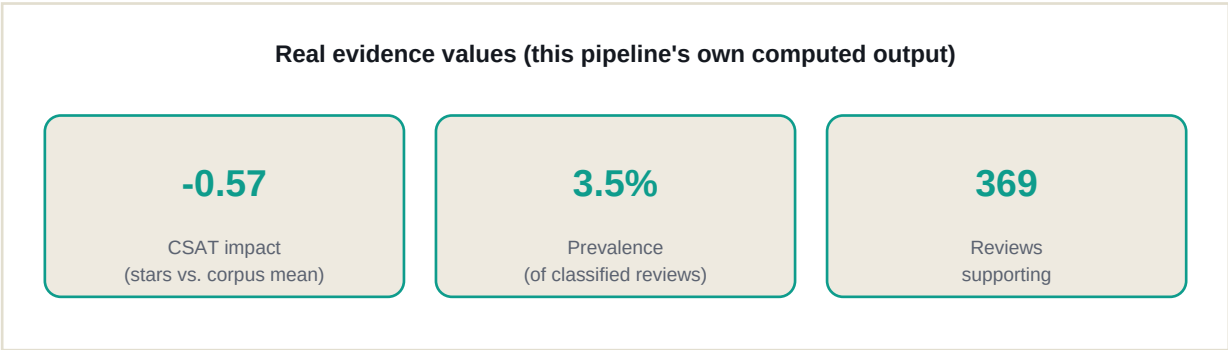


Fig. 2 — Real evidence figures computed by this pipeline for this friction theme.

4. Feasibility & Technical Risk

2-5yr feasibility rating	medium (rank 2 of 3 real candidates evaluated)
Feasibility rationale	No real trend document in this 12-document corpus directly addresses acoustic/noise engineering - TC-R11 (Dyson's sensor press release) is only tangential. Achievable, but without an external standard or precedent to build on, this requires proprietary acoustic R&D; from a colder start than the reliability theme.
Assumptions	- Acoustic engineering can meaningfully reduce PERCEIVED loudness without breaking the cost or power-draw envelope. - Reviewers who mention noise would actually change their rating if it were fixed, rather than noise being one complaint among several.

Known uncertainty

- Same price-coverage gap as OS-1.
- No direct WTP measurement exists for this or any theme (Q4).

5. Competitive Gap & Evaluation Record

Ambient-Sensing Night Purifier (ambient) — real white-space check: no real competitor product addresses this theme with this direction. Real competitor brands checked with no matching product: Dyson, Plixio, Oreck, Envion, Wynd, Molekule, AMEIFU, AZEUS, AIRTHEREAL, Egofine, Elechomes. Independent critic verdict (real, rule-based): **needs evidence**.

Noise-Cancelling Air Purifier (cross category transfer) — real white-space check: no real competitor product addresses this theme with this direction. Real competitor brands checked with no matching product: Dyson, Plixio, Oreck, Envion, Wynd, Molekule, AMEIFU, AZEUS, AIRTHEREAL, Egofine, Elechomes. Independent critic verdict (real, rule-based): **needs evidence**.

Pre-Sleep Air Purifier (temporal shift) — real white-space check: no real competitor product addresses this theme with this direction. Real competitor brands checked with no matching product: Dyson, Plixio, Oreck, Envion, Wynd, Molekule, AMEIFU, AZEUS, AIRTHEREAL, Egofine, Elechomes. Independent critic verdict (real, rule-based): **needs evidence**.

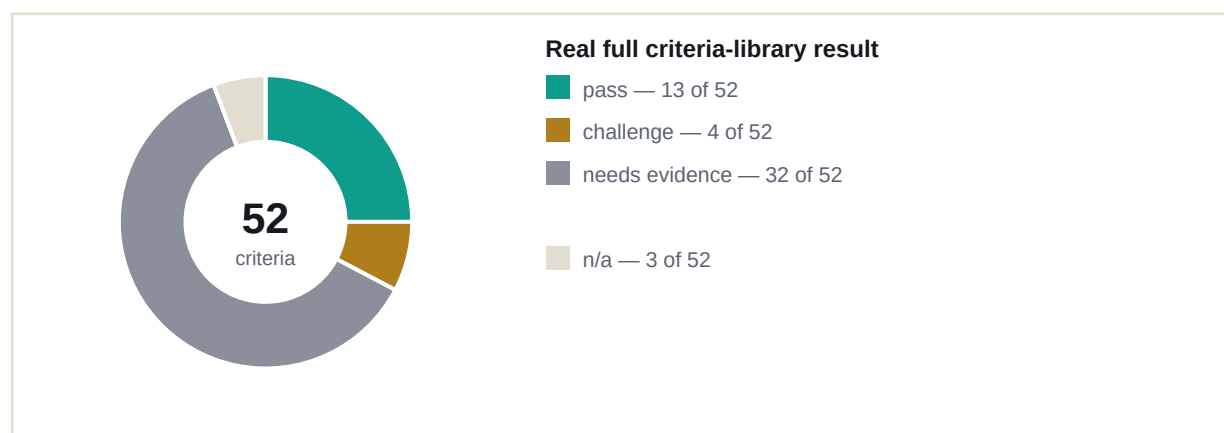


Fig. — Full real criteria-library result for "Ambient-Sensing Night Purifier" (representative direction for this theme; the same automated critic pass CriteriaWorld runs against every Magic Box concept, not a self-assessment by this disclosure).

Selected real criteria evaluations (evidence reality, hypothesis framing, and Versuni-specific capability checks):

ID	Status	Real note
E1	pass	McAuley-Lab Amazon-Reviews-2023 (Amazon.com real customer reviews) - real, dated (2004-10-14 to 2023-08-08).
H1	pass	Real friction theme: Noise / motor sound (weak real signal), n=369 reviews.
H2	pass	gate_passed=True
D2	needs evidence	category_assumptions.json::evidence_for_prevalence (real corpus-composition statistic, e.g. '72% of real corpus products').

V1	needs evidence	No real Versuni internal-capability dataset (portfolio inventory, HomeID/app install base, brand-permission research, IP/patent register, service network) exists anywhere in this pipeline - reported needs evidence, never assumed.
V2	needs evidence	No real Versuni internal-capability dataset (portfolio inventory, HomeID/app install base, brand-permission research, IP/patent register, service network) exists anywhere in this pipeline - reported needs evidence, never assumed.

6. Why This Real Outcome

This concept passed the real evidence-sufficiency gate — it is not being disclosed as a failure. Under the decision rule currently in force (pain_feasibility_majority), it was not selected: Non-dominated vs. OS-1, but the Pain + Feasibility majority rule favours OS-1.

Sensitivity. Switch to Economic Value override → picks Whisper-Quiet Night Mode instead.

7. What this doesn't cover yet

This pipeline's real evidence base has no physical-prototype, industrial-design, or manufacturing data of any kind — for this concept or for any real product in the underlying dataset (weight and dimension fields do not exist anywhere in the corpus, verified directly). So this disclosure leaves them open rather than guessing: physical weight and dimensions, bill of materials, housing material, the fabrication or manufacturing process, exact sensor part numbers, and firmware architecture. Filling any of these in without a real industrial-design phase would be invention, not evidence.

8. Design & Business Claims

The following claims describe the disclosed concept for internal evaluation purposes, generated from this candidate's own real evaluated fields, and are not filed with, or reviewed by, any patent office.

1. A connected household appliance addressing a real, evidence-derived friction: Motor/fan noise at higher speeds
2. The appliance of claim 1, intended for the real usage context: Bedroom, overnight use.
3. The appliance of claim 1, applying a design transform selected from the real catalogued directions for this friction theme: ambient (Explicit interaction -> invisible environmental behaviour.); cross category transfer (Transfer a verified capability from another product/category.); temporal shift (Move the job earlier/later.).
4. A method of evaluating such an appliance concept using this pipeline's real Consumer Pain evidence-sufficiency gate, real Economic Value computation, and real 2-5 year Feasibility rating, exactly as disclosed in this document.

9. Comparative Context

Real evidence values computed by this pipeline for every candidate evaluated in this funnel stage, for reference:

Candidate	Real prevalence	Real CSAT impact	Real typical price	Real status
OS-1 — Reliability-Verified Air Purifiers (exte	1.58%	-2.223	\$199.99	non dominated
OS-2 — Whisper-Quiet Night Mode	3.5%	-0.571	\$179.99	non dominated
OS-3 — Smart/Connected Feature Expansion (app	10.6%	no signal	no real figure for this	gate failed insufficient pain evidence

10. Evidence Appendix

Real evidence identifiers referenced in this disclosure: taxonomy:noise, TC-R11.

Underlying computation: `src/real/taxonomy_real.py` theme 'noise', n=369 reviews.

Market scenario used for category context: IMARC Group (Q5 alternative) (6.54% CAGR). The Q6 scores above do not depend on category CAGR at all - they are built entirely from review-level Consumer Pain/Economic Value/Feasibility, so this scenario flag changes the category-sizing narrative in Q5/insight_pack but changes NOTHING in this file's scores or verdict. Run with `--market-scenario=imarc` to see this confirmed live.